ReVive: an OpenCV 5 agent that restores old photos and asks when it isn't sure

OpenCV AI Competition 2026 (powered by AWS) · Agentic Vision with OpenCV 5 · Best Use of COOL
Repository: `github.com/samaunmahmud/revive` · Live demo: **[pending: CloudFront URL]**

> Status note (draft): sections marked **[pending]** need the AWS deployment and the COOL benchmark run on Graviton. Every other number in this report comes from scripts in the repository and can be reproduced (section 9).

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1. Summary

ReVive restores old, damaged photos (faded, tinted, grainy and scratched) the way a careful human restorer would. It **measures** what is wrong with OpenCV 5, **plans** one repair at a time, **acts** with an OpenCV 5 tool, and **re-measures** to check that the repair helped without doing harm. When a repair could change what the photo really shows, it **asks a person** instead of guessing.

On 35 seeded test cases (5 public-domain images × 7 damage types), ReVive raises mean SSIM against the clean original from 0.680 (damaged) to **0.822**, where a naive "run every filter once" pipeline drops it to 0.672. On undamaged photos ReVive applies no steps at all (**SSIM 1.000**), while the naive pipeline drops it to 0.75.

When a test planner injects 21 over-strong decisions, the verifier rejects the harmful steps and recovers almost all the lost quality: mean SSIM is 0.800 with the verifier off, **0.832** with it on, and 0.834 with no mistakes at all. Across 105 simulated LLM failures, every restoration still completed.

A person can also **correct the agent's perception**: paint damage it missed, or protect real detail it flagged. The agent applies the corrections at repair time and verifies them with its own measurement. One painted stroke repairs a wide scratch the detector cannot see (error inside it 118.6 → **14.7**, on a 0–255 scale), and fixes the 1942 real-photo miss.

2. Problem and users

Family photo archives, local-history societies and small museums hold millions of prints that are fading, yellowing and scratched. Restoration is skilled, slow work, and one-click "AI enhance" tools have the opposite problem: they apply every filter to every photo, and generative models can **invent** faces and details, which is unacceptable for historical records.

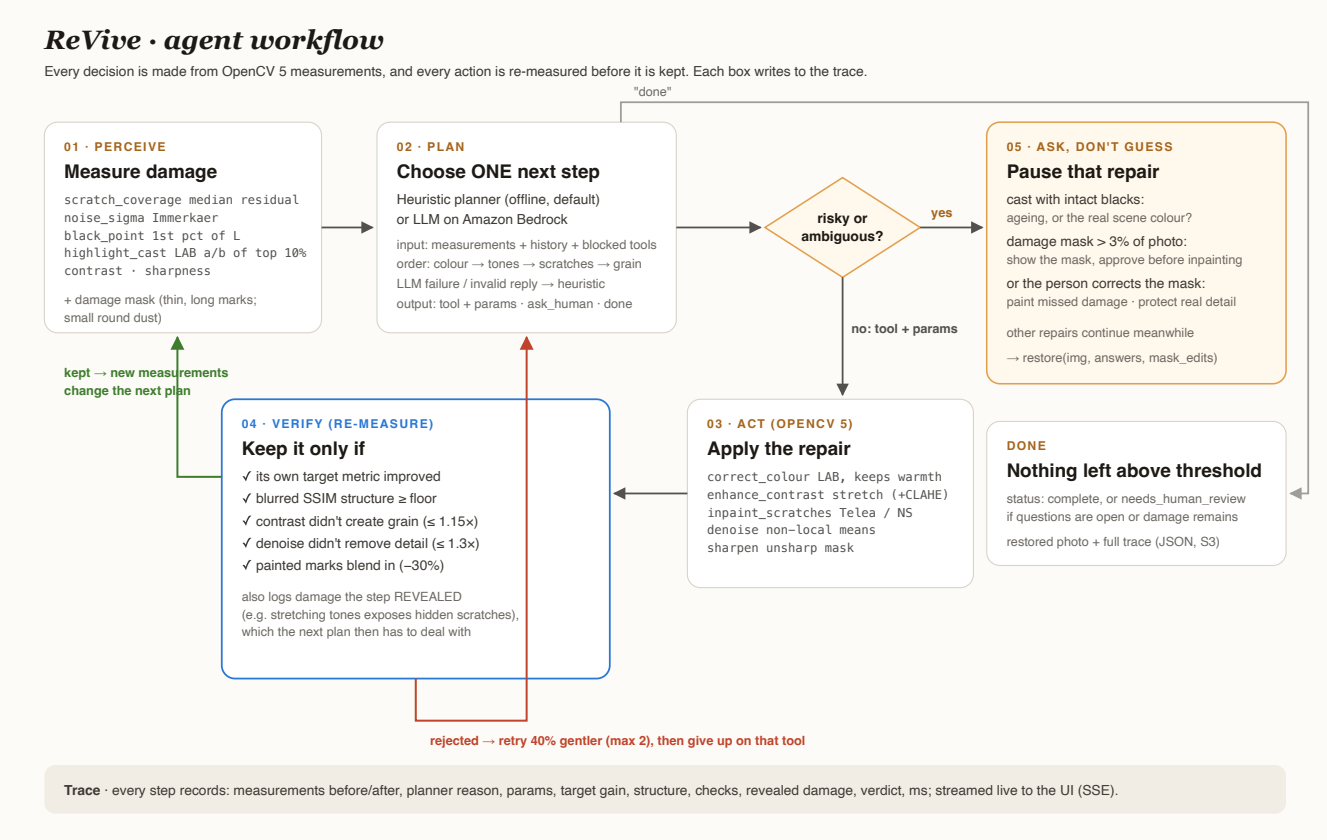
ReVive is for people who want old photos repaired **faithfully**:

- family historians digitising albums,
- archive volunteers working through boxes of prints,
- anyone who needs to know exactly what was changed and why.

Its design rules come from how a restorer works:

1. diagnose before treating;
2. do one thing at a time;
3. check every change;
4. never invent content without permission;
5. keep a record.

3. How the agent works



Perceive (`revive/perception.py`). Six measurements, all OpenCV 5:

measurement	method (OpenCV 5)	what it signals
`scratch_coverage`	`absdiff(gray, medianBlur(gray, 7))`, noise-scaled threshold, `connectedComponentsWithStats` shape rules (long and ≤ 2.5 px wide = scratch; small round = dust), `dilate`. Only **confident** marks count: those standing out ≥ 4.5× from the local texture (`blur`-based local std of the median image)	share of the photo covered by confident marks; the full mask is what gets repaired
`noise_sigma`	Immerkaer's estimator via `filter2D`	film grain
`black_point`	1st percentile of L in `cvtColor(BGR2LAB)`	fading (lifted blacks)
`highlight_cast`	LAB a/b offset of the brightest 10%	ageing tint
`contrast`, `sharpness`	L percentile range; `Laplacian` variance	tonal range, detail

Plan (`revive/agent.py`). A planner chooses **one** next step from these numbers: a tool with parameters, `ask_human`, or `done`.

- The **heuristic planner** is deterministic, works offline, and is the fallback. It follows a restorer's order: colour → tones → scratches → grain.
- The **Bedrock planner** sends the measurements, the step history and the list of blocked tools to Claude via the Amazon Bedrock Converse API, and parses a JSON action.
- The LLM **never sees pixels**, so every decision is grounded in OpenCV output.
- The LLM gets its rules in a system prompt generated from the same constants the heuristic and the verifier use, so the two planners cannot drift apart.
- Its replies are validated:
    - parameters are whitelisted and clamped to each tool's safe range, so an over-eager value can't crash or overdo a tool;
    - questions must name a real tool;
    - if the call fails, the reply isn't valid JSON, or it names an unknown or blocked tool, the heuristic decides that step instead, and the trace says so.

Act (`revive/tools.py`). Five OpenCV 5 repairs:

- `correct_colour`: LAB a/b shift that keeps a little natural warmth;
- `enhance_contrast`: percentile stretch on L, optional `createCLAHE`;
- `inpaint_scratches`: `inpaint` with `INPAINT_TELEA` or `INPAINT_NS`;
- `denoise`: `fastNlMeansDenoisingColored`;
- `sharpen`: unsharp mask with `GaussianBlur` and `addWeighted`.

Verify. After every action, perception runs again. A step is **kept** only if all of these hold:

1. its own target metric improved by a minimum amount;
2. structure is preserved: SSIM on blurred greyscale (`GaussianBlur`) stays above a per-tool floor;
3. it passes a side-effect check that needs no clean reference:
    - **contrast must not create grain:** noise-to-tonal-range may grow at most 1.15×. A pure fade-stretch measures about 1.0×; over-strong CLAHE measures 1.5–2.6×.

    - **denoise must not remove detail:** the removed signal ("method noise", after Buades et al.) may be at most 1.3× the estimated noise.

If any check fails, the step is discarded and retried 40% gentler (at most 2 retries), and then the agent gives up on that tool. The verifier also logs damage that a step **revealed**: stretching faded tones raises the measured grain and scratches, and later steps then repair them.

Ask, don't guess. Three situations become questions instead of actions:

- a **colour cast with intact blacks** might be the real scene (coffee, sunset), because aged prints fade their blacks as well;
- **damage covering more than 3%** of the photo would mean inventing large areas, so the person sees the red damage mask and approves it first;
- **residual damage** after automatic repair is flagged for review.

The other repairs continue while a question is pending. An answer re-runs the agent through `restore(img, answers={tool: bool})`.

Correct, don't just approve. For the scratch question, and for any finished result, the person can open a brush editor and correct the mask instead of only accepting or declining it:

- **paint** marks damage the detector missed: the agent repairs it even though its own measurements see nothing there, at the usual point in the order (after the tones, before the grain);
- **protect** marks real detail the detector flagged (thin poles, lettering, highlights): those pixels are never repainted.

The agent does not replay the edits blindly. It re-measures its own mask at repair time, which is after the tone step may have revealed more damage, and applies the corrections on top: (agent mask ∪ painted) – protected. The edits pass through `restore(img, answers, mask_edits={"add": …, "erase": …})`.

Painted marks need their own check, because the scratch detector may not see them at all; that is why they were painted. For those, the verifier measures how much the painted pixels stand out from their neighbourhood (mean lgray – 15×15 medianl, `perception.mark_anomaly`). After the fill this must drop by at least 30%, or the step is rejected and retried gentler. The verifier confirms that the fill blended in. It cannot judge whether the person was right to paint there; that decision is theirs, and it is recorded in the trace.

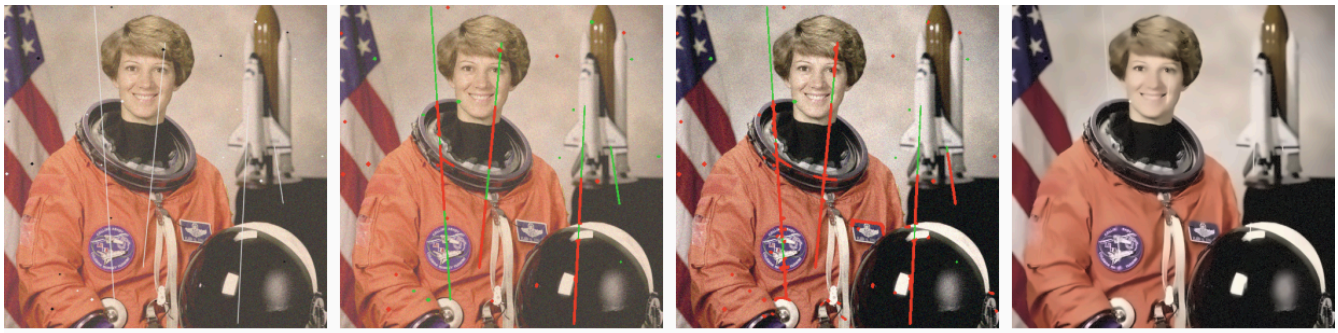
Trace. Every step records:

- the measurements before and after,
- the planner's reason and parameters,
- the target gain, structure score and check values,
- any revealed damage,
- the verdict and the time taken.

The web UI streams these live, and the trace is downloadable as JSON.

4. OpenCV output changes the agent's decisions

`docs/trace-example.md` annotates three real traces. The clearest one is below.



1. input: tinted, faded, grainy, scratched | 2. damage found at start: 54% of true marks | 3. after tone step: 78% of true marks | 4. after inpainting + denoise

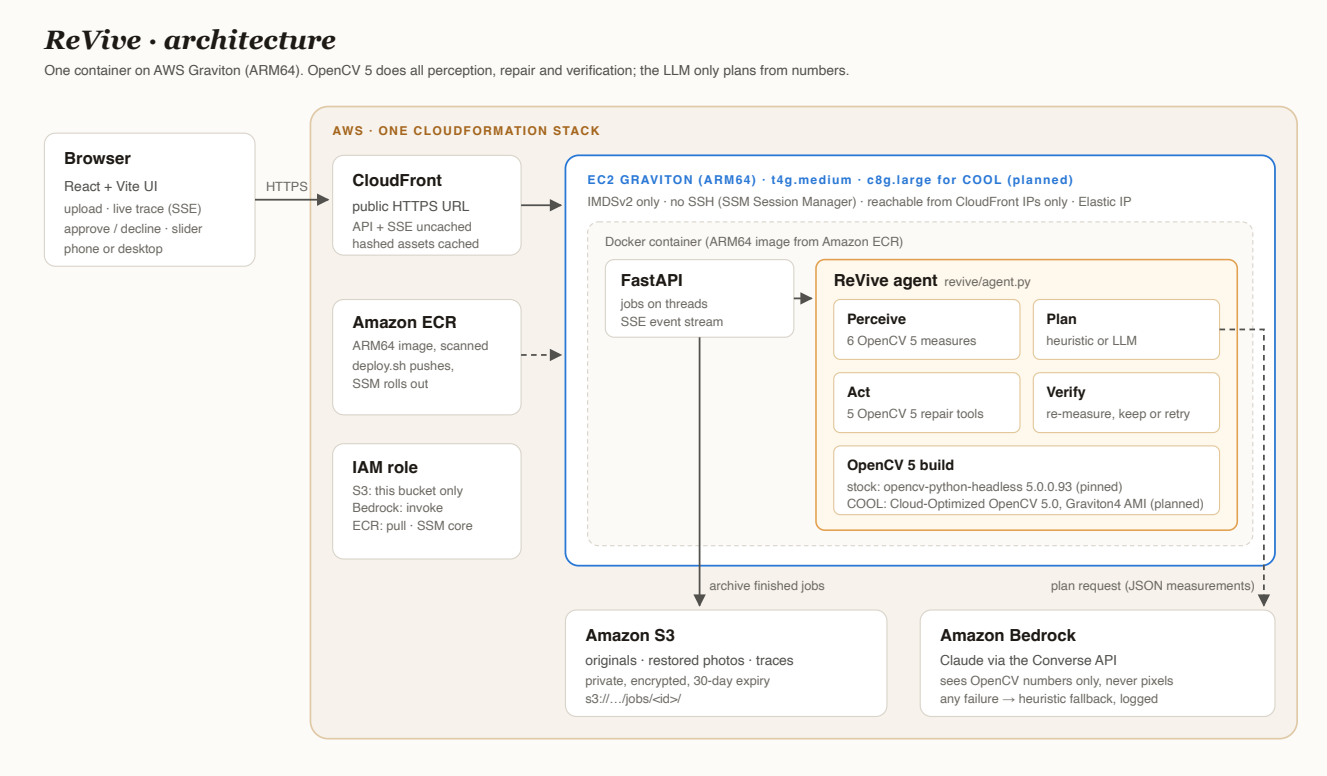
red = mask ReVive measured with OpenCV 5 green = true scratch/dust positions (known because the damage is synthetic)

On the "Faded portrait" sample, the scratch mask measured at the start finds **54%** of the true scratch and dust pixels (precision 96%). These are known exactly, because the damage is seeded. After the tone-restoration step, the re-measured mask finds **78%** (precision 79%). Inpainting therefore uses the new mask, and the denoise strength (10.1) is set from the re-measured grain, not the starting 7.1. A measure-once pipeline would have missed nearly half the scratches.

The other two examples:

- **A rejected step:** over-strong CLAHE is rejected because it created grain (1.56×), and the gentler retry is kept.
- **A question instead of an action:** a measured cast of 21.9 with a black point of 0.012 turns into a question rather than a colour correction.

5. Architecture and AWS deployment



component	role
EC2 Graviton (ARM64)	runs one Docker container: FastAPI, the React UI and the agent. t4g.medium (4 GB: a 16 MP full-size restore peaks at ~2.1 GB, measured); c8g.large with COOL **[pending]**
CloudFront	public HTTPS URL. API and Server-Sent Events uncached; hashed assets cached at the edge
Amazon Bedrock	LLM planner (Claude, Converse API)
Amazon S3	archive of every finished job (original, restored photo, trace): private, encrypted, 30-day expiry
Amazon ECR	ARM64 image, scanned on push
IAM / SSM	least-privilege instance role; administration through SSM Session Manager

Security.

- The instance accepts HTTP only from CloudFront's managed prefix list.
- IMDSv2 is required.
- There is no SSH.
- Uploads are size-checked (15 MB, 40 MP), decoded with `imdecode` (EXIF rotation, 16-bit TIFF, greyscale and transparency handled), and restored live as a 1600 px preview.
- **Full size:** "Restore at full size" runs the agent again on the upload (up to 4096 px) with the same answers. It does not replay the preview's steps, because downscaling averages away fine grain and thin scratches. Replaying a 4× downscaled plan at full size left the grain in (SSIM 0.36), while re-measuring at full size reached 0.75. On a 2400 px upload, the full-size run found and removed grain the preview could not see.

Deployment.

- `deploy/deploy.sh` builds the ARM64 image, pushes it to ECR, and creates or updates the single CloudFormation stack.
- Later releases roll out through SSM Run Command, without replacing the instance.
- `deploy/teardown.sh` removes everything.
- Running cost is about **$0.95 a day** (t4g.medium).
- Full-size runs go through a single slot, so only one heavy job runs at a time.

Deployment results [pending]: live URL, Bedrock model and region that work on the account, end-to-end latency through CloudFront.

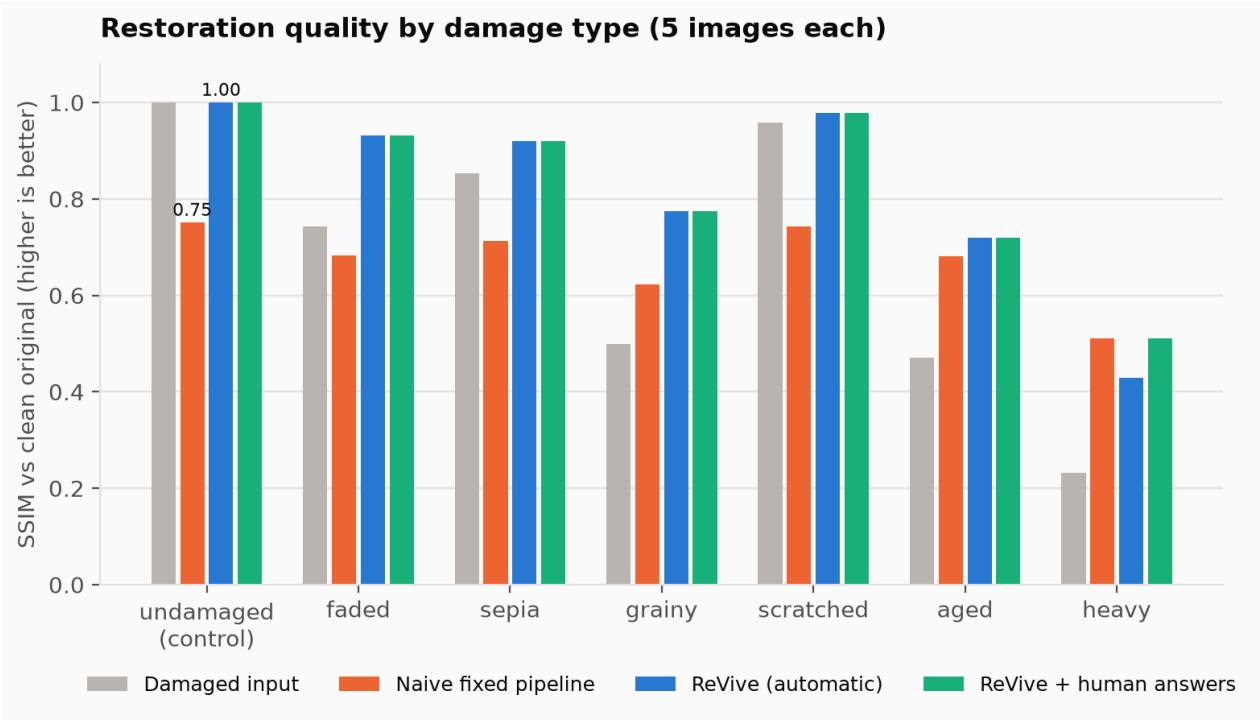
6. Evaluation

Setup.

- **Images:** 5 public-domain photos from scikit-image (astronaut, coffee, cat, rocket, cameraman).
- **Damage profiles:** 7 seeded ones (none, faded, sepia, grainy, scratched, aged, heavy).
- **Scoring:** PSNR and SSIM against the clean original.
- **Baselines:** the damaged input, and a naive fixed pipeline that runs every tool once at default settings with no verification.
- **Simulated human:** answers truthfully. It approves masks, and says "ageing" only when the profile really added a tint.

6.1 Restoration quality

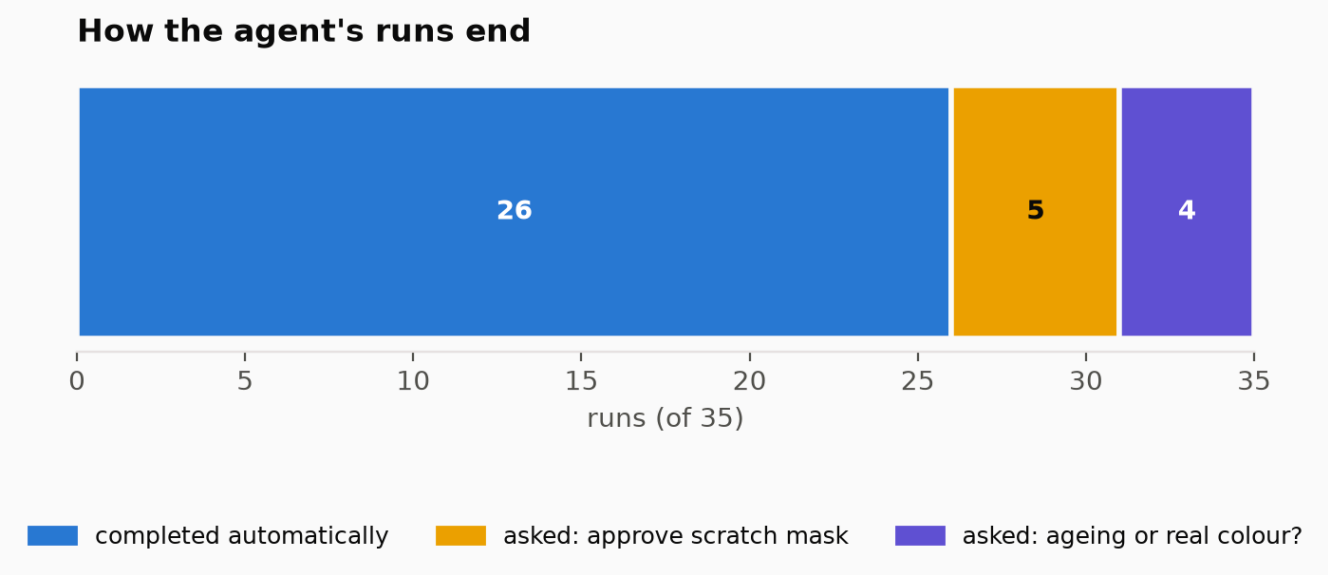
(35 cases)	SSIM	PSNR (dB)
damaged input	0.680	25.1
naive fixed pipeline	0.672	18.8
ReVive, automatic	**0.822**	**29.4**
ReVive + human answers	0.834	29.5



ReVive wins clearly on every damage type except **heavy**. There the naive pipeline (0.51) beats automatic ReVive (0.43), and ReVive only ties it with human approval (0.51). The agent refuses to inpaint about 10% of the photo on its own, which is by design. Even approved, its gentler repairs leave more damage than an aggressive filter chain. On **undamaged** photos ReVive applies nothing at all (1.000), while the naive pipeline does visible harm (0.75).

6.2 Agentic behaviour (`eval/agentic.py`)

metric	value
runs completed automatically	26 / 35 (74%)
escalated to a human	9 / 35 (5 mask approvals, 4 colour questions)
no harm (SSIM not lower than the damaged input)	97%
task success (no harm and final damage score ≤ 1)	83%
runs where re-measurement found more damage after a step	19 / 35 (see 7)
inpaint steps on photos with no scratches	0 of 10
steps kept / rejected (normal planner)	61 / 0

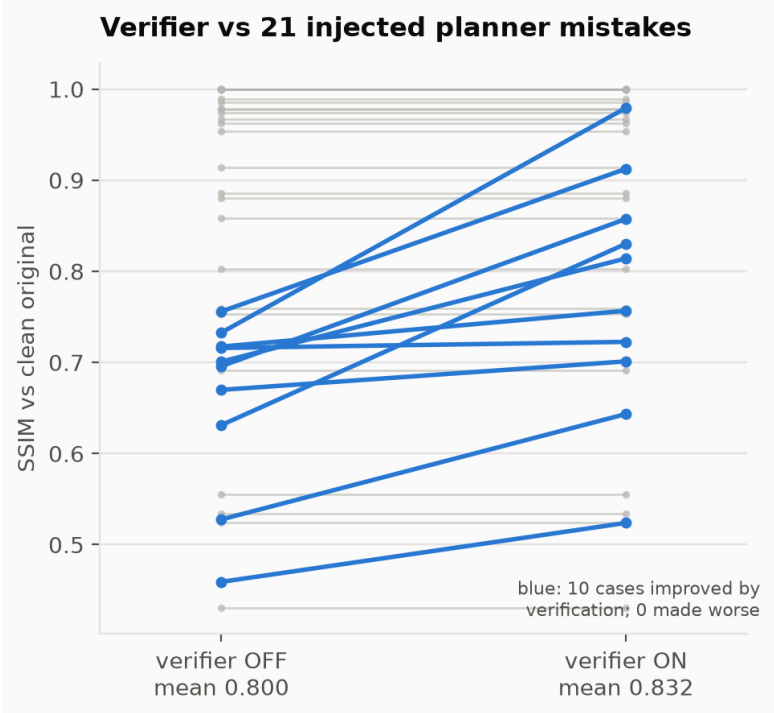


Failure handling 1: a planner that makes mistakes. A test planner replaced the settings with over-strong ones on half of its first attempts:

- denoise strength ×3;
- CLAHE clip 4;
- inpaint radius 9 with Navier-Stokes;
- full-strength colour correction.

It injected 21 mistakes over the same 35 cases.

	mean SSIM	photos not harmed
verifier off	0.800	91.4%
verifier on	**0.832**	**97.1%**
reference: no mistakes	0.834	97.1%



- The verifier rejected 12 steps, and every gentler retry succeeded.
- It improved 10 cases and made none worse.
- These checks were added *because* this experiment exposed a gap. The first version of the verifier (target improved + structure) accepted every injected mistake, because each one improves its own metric. We report that openly.

Failure handling 2: the LLM fails. The real `BedrockPlanner` was run against a fake client that returned broken JSON, prose, unknown tools and blocked tools, or timed out (105 failures across 35 runs). All 35 runs completed. Every result was pixel-identical to the heuristic planner's, and every fallback is labelled in the trace.

Observability. Every decision is in the trace, with:

- the measured evidence and the planner's reason;
- parameters, target gain, structure and check values;
- revealed damage, the verdict and the time in ms.

The UI streams it live, and the JSON is stored in S3.

Human control.

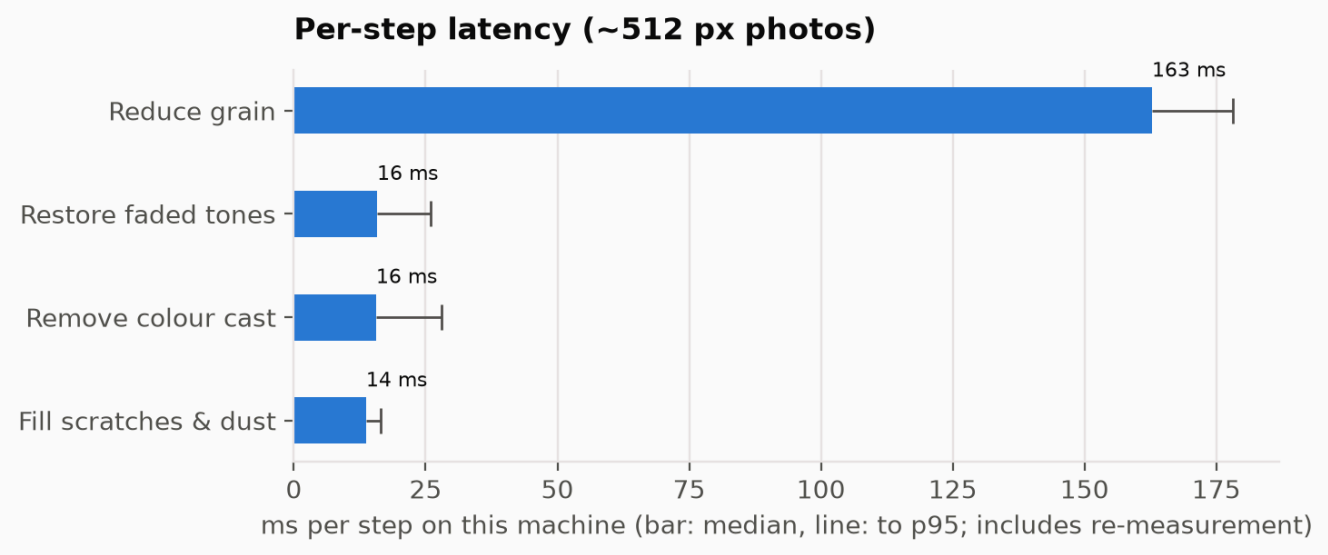
- Risky repairs pause for approval, with the evidence shown (the red mask).
- A "decline" blocks that tool, and overrides any painted marks.
- The person can correct the mask itself (paint missed damage, protect detail); see 6.8 for measurements.
- Answers and mask edits (share of the photo painted and protected) are recorded in the trace, and every step that used them is badged "your mask" in the UI.

Latency (median per step on the evaluation photos, ~512 px, including re-measurement):

step	median latency
colour	16 ms
tones	16 ms
inpainting	14 ms
denoise	163 ms (p95 178 ms)

Measured on an Apple M-series laptop; an earlier run on the same machine under load measured 2–3× higher. The Graviton numbers are in 6.4.

Non-local-means denoising dominates the run time.



6.3 Bedrock LLM planner vs heuristic [pending]

`eval/planners.py` runs both planners on the same cases through the same verifier. It records quality, escalations, fallbacks, and per-call latency and tokens.

6.4 COOL vs stock OpenCV 5 on Graviton [pending]

`bench/benchmark.py` measures per-step latency, full-restore latency, throughput and cost per 1,000 restores. `deploy/cool_bench.sh` runs it end to end on AWS with one command:

- pricing comes from the AWS Pricing API;
- each instance self-terminates;
- a run is only labelled COOL if `cv2` is loaded from the COOL install. It runs with both OpenCV builds on the same Graviton4
  instance (c8g.large), plus an x86 reference. COOL is the Cloud-Optimized OpenCV 5.0 Graviton4 AMI; stock is pip `opencv-python-headless==5.0.0.93`.

A note before the numbers: stock OpenCV 5 wheels already include KleidiCV, and denoising dominates ReVive's runtime. So any COOL gain depends on how well it accelerates `fastNlMeansDenoisingColored`, and we will report whatever we measure.

6.5 Fixing false scratch detections

The first full evaluation showed that thin real detail was being flagged as scratches: 9 of 19 kept inpainting steps were on photos with no marks (a launch tower lattice, tripod legs, a cup rim). We separated two questions:

- **whether** to repair, decided from *confident* marks only;
- **what** to repair, using the full mask, so recall is unchanged.

A mark is confident when it stands out strongly from the texture right around it. Per pixel, that is `|image − median| / (local std of the median image + 2)`, averaged over the mark and required to be ≥ 4.5.

	before	after
mean SSIM, automatic / with human	0.818 / 0.830	**0.822 / 0.834**
mean PSNR, automatic	27.3 dB	**29.4 dB**
undamaged control	0.994	**1.000**
inpaint steps on photos with no marks	9 of 19	**0 of 10**
aged/scratched cases still repaired	10 / 10	10 / 10

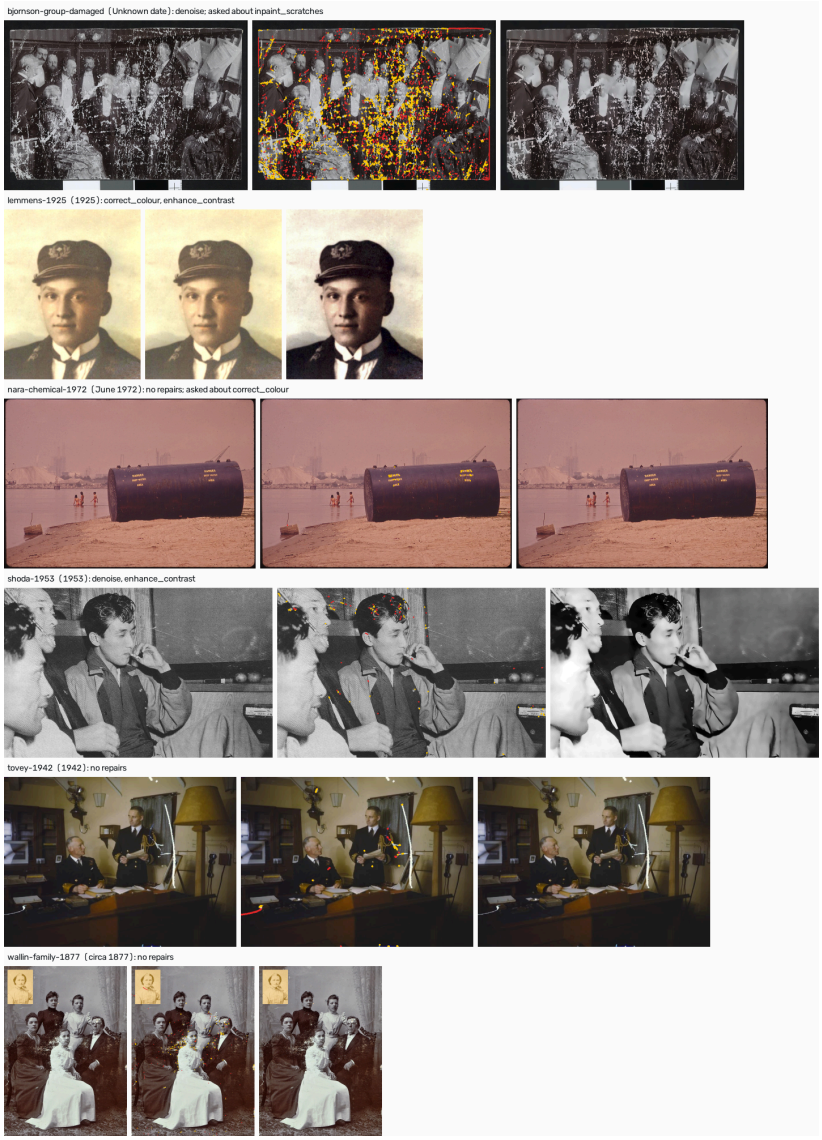
Our first version measured contrast against a ring around the whole mark. That hid long scratches that cross many regions, and the flagship demo photo stopped being repaired. Measuring per pixel fixed it. Cutoffs from 4.0 to 5.0 give identical results, and 6.0 starts missing real scratches, so 4.5 sits in the middle of that plateau.

6.6 What we tried that did not work

Joining scratch fragments with `HoughLinesP`. The idea was to recover scratches that break up over similar tones. It found texture on clean photos (control SSIM 0.994 → 0.991), and it slightly lowered the aged cases. It only gained +0.004 on heavy damage, so it was not shipped.

6.7 Real photos (qualitative)

There is no clean original for a real damaged photo, so these are judged by eye. `eval/real_photos.py` runs the agent on six public-domain scans from Wikimedia Commons (sources and licences in `eval/real_photos/credits.json`).



Columns: input · repair mask (yellow = full mask, red = confident marks) · automatic result.

photo	what ReVive did	assessment
portrait, 1925 (faded, tinted)	colour cast removed, then tones restored	**good:** neutral tones, real blacks back
colour slide, 1972 (magenta dye fade)	asked "ageing or real colour?" because the blacks are intact; after "ageing", cast 25.5 → 9.2	**good:** the caution is right, and the approved result looks natural
group photo with heavy emulsion loss	asked before repainting 5.7% of the photo	**correct to ask.** The losses are too large for thin-scratch inpainting either way
naval officers, 1942 (one wide, curved white scratch)	nothing	**miss:** the scratch is 5–8 px wide (the detector expects ≤ 2.5 px), and one scratch is too small a share of the photo to trigger a repair. **Fixed with one painted stroke (6.8).**
family, 1877 (scattered emulsion flecks)	nothing	**under-repaired:** few flecks pass as confident marks
1953 press photo (coarse print grain)	denoised (strength 15), tones restored	**over-smoothed:** faces and wood look waxy, even though the method-noise check passed, because the grain really was that strong

The real photos also exposed a **safety bug**, now fixed. After the confident-marks change, the 3% approval rule compared the confident subset (2.2%) while inpainting repainted the full mask (5.7%), and it did so without asking. The rule now uses `repair_area`, the share of the photo that would actually be repainted.

We also tried detecting wide scratches with an image pyramid (½ and ¼ scale). It lowered the synthetic scores (0.822 → 0.818) and still missed the 1942 scratch, so it was not shipped.

6.8 Human mask corrections (`eval/human_edits.py`)

1. Damage the detector misses. Each test photo gets the "scratched" profile plus one 6 px scratch, wider than the detector's 2.5 px limit. A person paints it with a 9 px stroke. Error is measured against the clean original inside the scratch (mean absolute difference, 0–255).

image	detector found	damaged	automatic	painted	SSIM auto → painted
astronaut	0.5%	100.7	100.3	7.9	0.961 → 0.973
coffee	2.0%	123.5	122.3	21.8	0.948 → 0.956
chelsea	1.8%	122.5	121.2	20.7	0.962 → 0.974
rocket	1.0%	169.4	168.5	2.5	0.971 → 0.988
camera_bw	1.0%	80.8	80.7	20.4	0.956 → 0.965
mean		**119.4**	**118.6**	**14.7**	**0.960 → 0.971**

The verifier kept all 5 painted repairs. Brush width matters: on the same scratches an 8 px brush leaves an error of 13, and a 24 px brush 28, because a wide brush repaints clean pixels too. The editor therefore starts with a thin brush and tells the person to paint just over the mark.

2. Real detail flagged as damage. On the "scratched" profile the true marks are known (damaged – clean), so parts of the agent's repair mask more than 3 px from any real mark are false detections: helmet rims, the shuttle's tank edge, badge and lip edges. A simulated person protects them.

image	false detections	error on them: automatic	protected	SSIM auto → protected
astronaut	0.40%	66.2	0.0	0.978 → 0.985
coffee	0.77%	69.0	0.0	0.963 → 0.980
rocket	0.26%	45.7	0.0	0.989 → 0.992
camera_bw	0.73%	64.2	0.0	0.974 → 0.985

(chelsea had no false detections.) Protected pixels are never repainted, so their error is exactly the damage they already had, here none. This is an upper bound: a real person protects only what they notice.

3. The real scan the agent missed (1942). The detector covers 0.6% of the wide curved scratch, so the automatic run changes nothing. With one painted stroke along it, the agent repairs it and the verifier measures the stroke's anomaly falling 57.8 → 9.5 (structure 0.988).



Input · the stroke a person paints · automatic (missed) · with the painted stroke.
Where the scratch crosses the picture-frame edge the fill smears slightly:
Telea inpainting propagates the nearest surrounding colours.

7. Limitations

- **The scratch cutoff is tuned on synthetic marks.** Our seeded scratches are high-contrast. Faint real scratches in busy texture may not count as confident, and then they are not repaired. Real prints are needed to check this.
- **"Revealed" means "measured higher after this step"**, not "verified hidden damage". 6 of the 8 "revealed scratches" events happen on photos with no marks. Since 6.5 they no longer trigger a repair, but the badge still appears in the trace.

- **Painted fills are classical inpainting.** A wide painted region is filled from its border, so long or wide strokes across strong edges can smear (6.8). The verifier checks that the fill blends in, not that it is historically right.
- **Denoising softens texture** (wood grain, fur) while still passing the verifier.
- **Heavy damage:** the conservative agent restores less than an aggressive filter chain, even with approval.
- **Warm subjects under a tint** are slightly over-corrected.
- **Real scratches wider than ~3 px are missed,** and so is a single scratch on an otherwise clean photo (6.7). The detector is built for thin film-style scratches.
- **Strong real grain is over-smoothed** (6.7). A grain-retention option would suit archival work better.
- **Quantitative results are on synthetic damage.** Real photos are evaluated qualitatively only, because there is no clean original to compare against.
- **Overfitting risk.** The verifier limits (1.15×, 1.3×) and the perception thresholds were tuned on the same five images we evaluate on. A held-out set of real prints is needed.

8. Responsible use

- **ReVive repairs; it does not invent.** It uses classical OpenCV operations only. There is no generative fill and no face synthesis.
- **Large-area inpainting and ambiguous colour changes need human approval.**
- **Every change is logged with its evidence,** and the trace travels with the result.
- **Restored images should be labelled as restored** when used as historical records. The trace JSON documents what was changed.
- **Uploads are processed privately** and archived in a private S3 bucket that deletes them after 30 days.
- **No photo is used for training.**
- **The LLM receives six numbers and a step history, never the image.**

9. Reproducing the results

```
pip install -r requirements.txt
python -m eval.evaluate        # quality table (6.1)
python -m eval.agentic         # agentic metrics, verifier ablation, LLM-failure test (6.2)
python -m eval.trace_example   # annotated trace and ground-truth mask numbers (4)
python -m eval.real_photos     # real public-domain scans: contact sheet (6.7)
python -m eval.human_edits     # painted / protected mask corrections (6.8)
python -m eval.charts          # figures in docs/figures
python -m eval.planners        # Bedrock vs heuristic (needs AWS credentials)
python -m bench.benchmark run --label stock   # COOL benchmark (run on Graviton)
```

Pinned dependencies are in `requirements.txt`; the ARM64 container is `Dockerfile`.